



# User Experience Management in Smart Washroom

Company: ComA/ComB

## SYNOPSIS

The project leverages on IoT and emerging web technologies to better manage washrooms, thus, enhancing the overall washroom user's experience.

## CONCEIVE

Solution must be customizable for flexible selection by clients, allow 24/7 monitoring of toilet, provide instant notification, modular in design and highly scalable for easy deployment.

## DESIGN

**Software:** XAMPP, Charting Library, Cloud computing services, Integrated development environment (IDE).

**Hardware:** Sensors, Sensor housings, Microcontrollers, Router, Batteries.

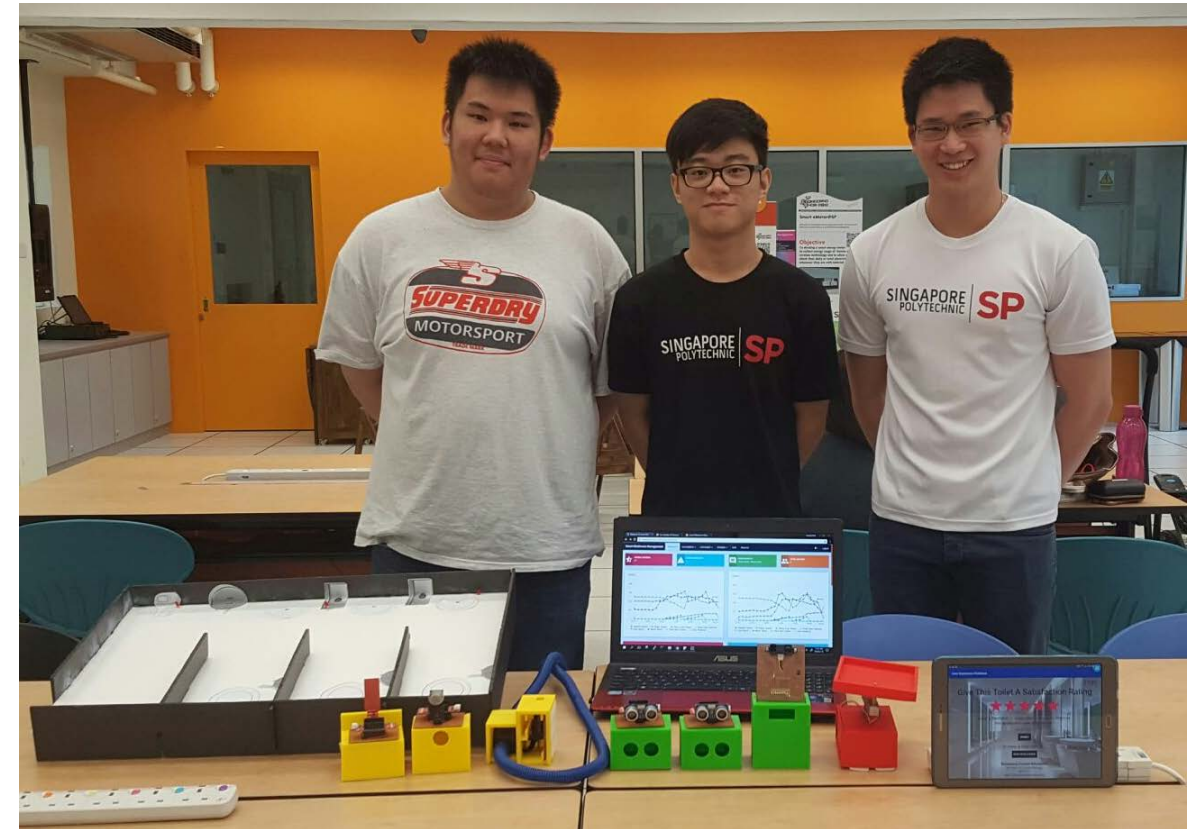
## IMPLEMENT

**Stages:** Source, design, create and integrate sensors, build server and client-side web portal.

**Each stage involves:** Prototyping, testing, troubleshooting and improving product.

Company A

Company B



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Company Liaison officer name: Mr. Liow Wei Quan (NCS), Ms. Sharon Liao (Singtel)

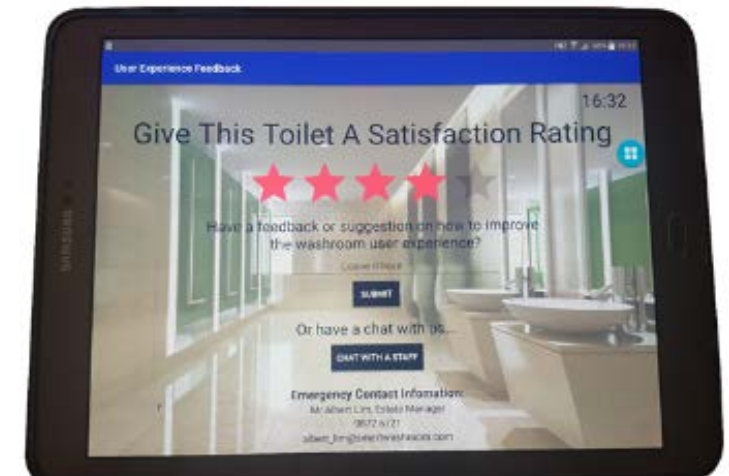
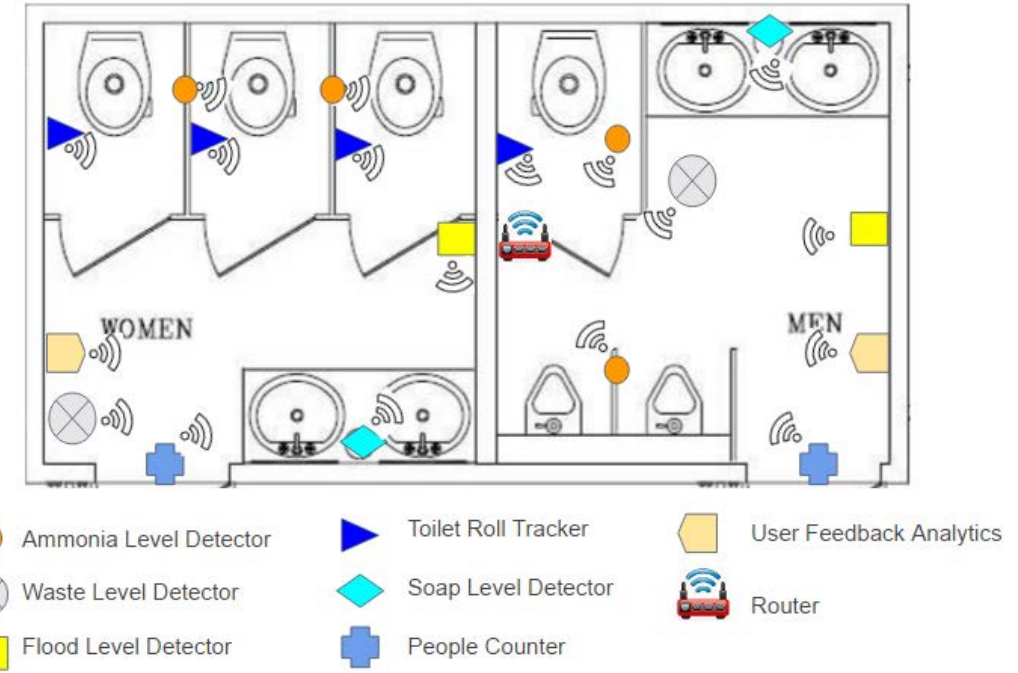
# CONCEIVE

## What are the users' needs

Product must be customizable for flexible selection by clients, allow 24/7 monitoring of toilet, provide instant notification, modular in design and highly scalable for easy deployment.

## What features will help meet these needs to achieve the functional goals of the project?

- Seven different sensor modules.
- A User Feedback Application.
- A web portal with differentiated logins for managers and cleaners.
- Real-time monitoring of updates,
- Real-time data visualisation and data analytics,
- Aggregated data sorted by Day, Week and Month,
- Display of calculated data (e.g. peak and average),
- Notifications on web portal and mobile devices,
- Chat messaging system,
- Online scheduling system.



# Brainstorming Ideas

Are our ideas innovative and up to industry standards?

# Which **Cloud Computing Services** (AWS vs. Azure vs. Google Cloud vs. IBM) should we choose?

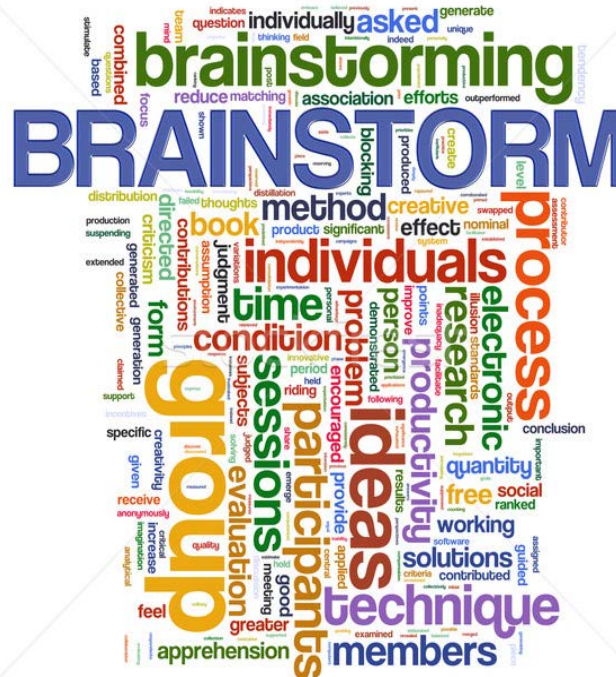
Which **Wireless Communication Protocol** (ZigBee vs. WiFi, Bluetooth Smart) is most suitable?

# Which **Microcontroller** (ESP8266 vs. Arduino vs. Raspberry Pi) is the most suitable?

Who are our **Direct and Indirect competitors** out there?

Which **Data Interchange Format** (JSON vs. XML) is most suitable?

# Which **Data Visualisation Software** (Tableau vs. Splunk) is most suitable?



# DESIGN

## Choice of hardware and/or software, including standards

### Hardware:

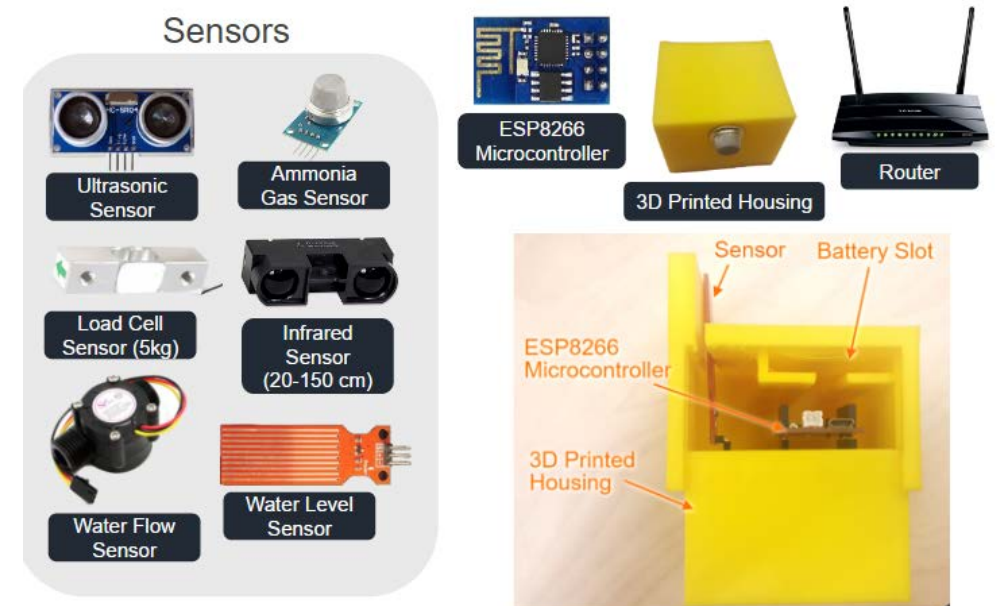
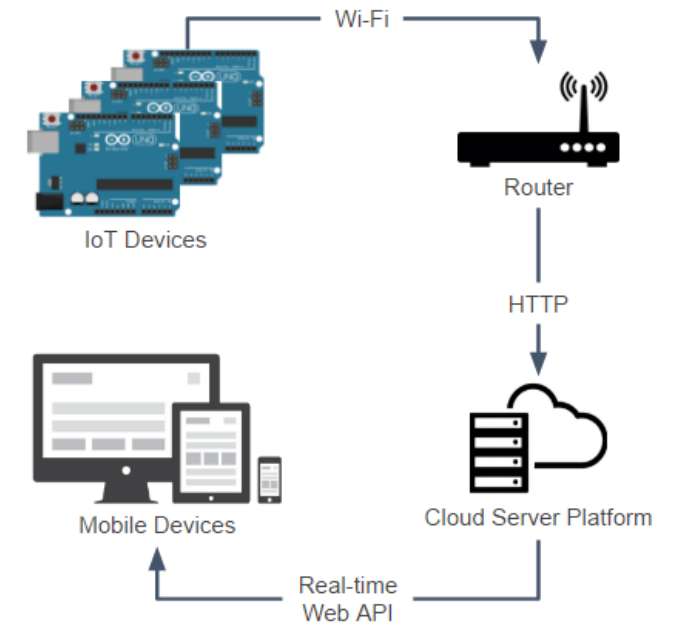
1. Sensors (ultrasonic, water level, load cell, infrared, ammonia gas, water flow),
2. ESP8266 microcontroller,
3. 3D printed housing,
4. Router and
5. Android tablet.

### Software:

1. XAMPP on Windows (Apache Web Server & MySQL Database),
2. Highcharts (JavaScript Charting Library),
3. Firebase Cloud Messaging,
4. Heroku,
5. Android Studio IDE and
6. Arduino IDE.

## Systems Integration (How does the above work together?)

1. Microcontroller reads sensor data and sends to Web Server via Wi-Fi,
2. Code on Web Server updates Database,
3. Various integrated platforms access data in Database to generate charts for data visualisation, and perform data analytics to summarise trends and activate notifications, where applicable.

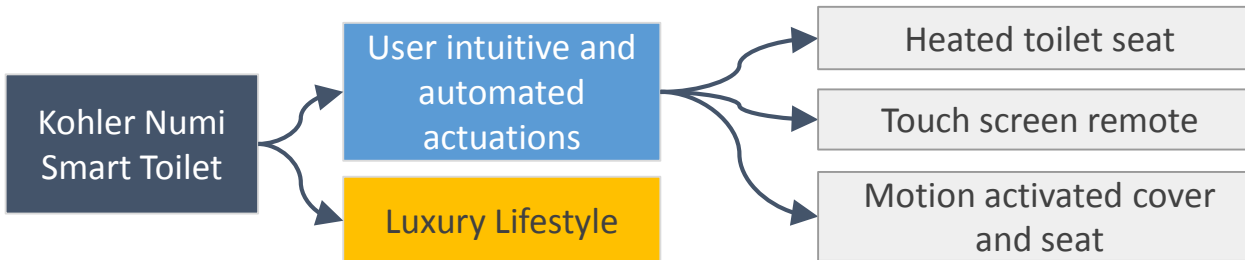




# Market Analysis



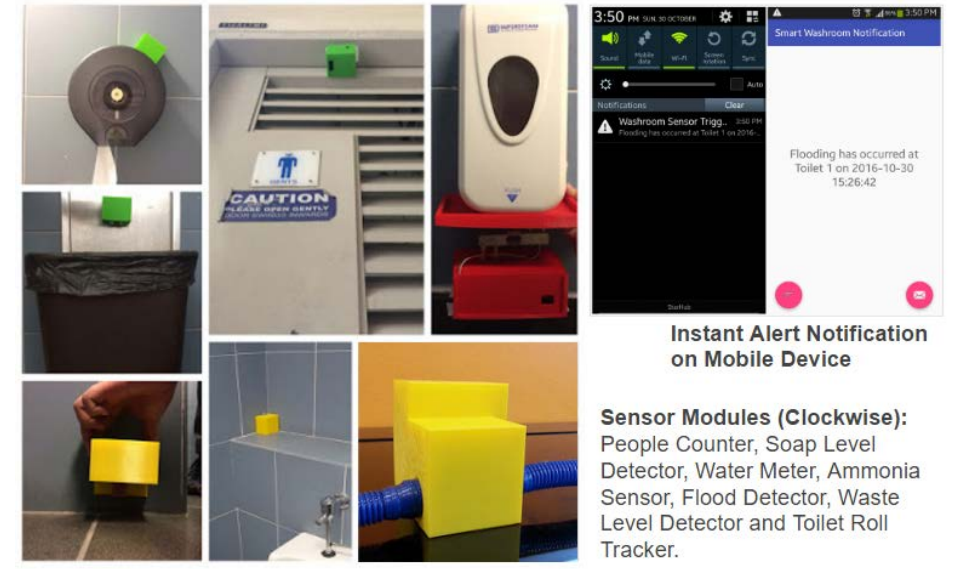
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# IMPLEMENT

## List of activities carried out in implement stage

1. Sourcing of hardware components,
2. Assembly and testing of hardware components,
3. Setting up of XAMPP and web portal,
4. Integration of database, web portal and sensor modules,
5. Design of graphs and data analysis,
6. Programming and further integration of differentiated login, chat messaging system, email & mobile notification and online scheduling system,
7. Troubleshooting and simulation testing,
8. Implement improvements based on user and industry feedback,
9. Demo and presentation of prototype,
10. Preparation of report and instruction manual.



| sensor_id          | value  | timestamp           | location |
|--------------------|--------|---------------------|----------|
| WasteLevelDetector | 100.00 | 2016-11-30 08:48:55 | Toilet 1 |
| WasteLevelDetector | 100.00 | 2016-11-30 08:49:36 | Toilet 1 |
| AmmoniaSensor      | 410.00 | 2016-10-31 00:17:30 | Toilet 1 |
| AmmoniaSensor      | 409.00 | 2016-10-31 01:17:30 | Toilet 1 |
| AmmoniaSensor      | 409.00 | 2016-10-31 02:17:30 | Toilet 1 |
| AmmoniaSensor      | 410.00 | 2016-10-31 03:17:30 | Toilet 1 |
| AmmoniaSensor      | 408.00 | 2016-10-31 04:17:30 | Toilet 1 |
| AmmoniaSensor      | 400.00 | 2016-10-31 05:17:30 | Toilet 1 |
| AmmoniaSensor      | 420.00 | 2016-10-31 06:17:30 | Toilet 1 |
| AmmoniaSensor      | 430.00 | 2016-10-31 07:17:30 | Toilet 1 |
| AmmoniaSensor      | 440.00 | 2016-10-31 08:17:30 | Toilet 1 |
| AmmoniaSensor      | 455.00 | 2016-10-31 09:17:30 | Toilet 1 |
| AmmoniaSensor      | 500.00 | 2016-10-31 10:17:30 | Toilet 1 |
| AmmoniaSensor      | 550.00 | 2016-10-31 11:17:30 | Toilet 1 |
| AmmoniaSensor      | 580.00 | 2016-10-31 12:17:30 | Toilet 1 |
| AmmoniaSensor      | 600.00 | 2016-10-31 13:17:30 | Toilet 1 |
| AmmoniaSensor      | 650.00 | 2016-10-31 14:17:30 | Toilet 1 |
| AmmoniaSensor      | 410.00 | 2016-10-31 15:17:30 | Toilet 1 |
| AmmoniaSensor      | 440.00 | 2016-10-31 16:17:30 | Toilet 1 |
| AmmoniaSensor      | 450.00 | 2016-10-31 17:17:30 | Toilet 1 |

# BENEFITS

## How has the project benefit the users?

- Modular approach allows sensor modules to be **mixed and matched** according to the needs of a particular washroom. The set-up can be extended even beyond washrooms.
- Live monitoring of the washrooms via the sensor modules **eliminates the need for manual checks**.
- Insights gained on usage patterns result in more **effective deployment of cleaners**.
- Auto-alerts and notifications allow more **efficient response to emergencies**.
- Synchronous chats between supervisor and cleaners result in **better communications**.

Overall higher standards of hygiene and better experience for users.

